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(1) Research Program/Project/Study Title: DEVELOPMENT OF LOW SUGAR READY-TO-DRINK CALAMANSI JUICE	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	6
(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

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(7) Sector/Commodity
Calamansi

(8) Discipline/Field of Specialization
Food Science and Technology

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

One of the major health issue confronting our world today is the global increase in the rate of overweight and obesity. While Philippines is battling the persistent problem of undernutrition, particularly underweight, stunting, wasting and chronic energy deficiency (CED), it also faces the burden of overweight and obesity. Report of FNRI (2018) declares a rising prevalence of overweight and obesity across different age group, affecting 18 million Filipinos according to ASEAN survey in 2016.

The increasing cases of obesity and overweight is found to be associated with the consumption of energy-dense foods, specifically foods rich in sugar and fats (WHO, 2020). Sugar, one of the most universally used food ingredients, has been implicated to several negative health consequences. Result of several studies found an association between high sugar intake and increased risk of obesity such as hypertension, dyslipidemia, diabetes, insulin resistance and other diet-related health consequences (Morenga, Mallard, & Mann, 2013). As such, sugar sweetened beverages including soft drink and fruit juices is considered to be a leading source of added sugar in the human diet and reformulation of such product to enhance its health value is a good option.

Stevia have attracted an increasing attention as a natural sweetener and sugar substitute derived from the leaves of the plant specie *Stevia rebaudiana*. It is claimed to have a relative sweetness of 250-300X as compared to sucrose. A cited in the study of Alizadeh (2014), it was first commercialized as a sweetener in 1971 and are recognized as safe supplements by JECFA, WHO and Food and Drug Administration (FDA) with relatively high upper limits and in the year 2008, it was pronounced as GRAS or "Generally Recognized as Safe" food ingredient.

CapSU Pontevedra has a growing calamansi project which has already started its sale of fresh calamansi fruits to consumers. In addition, the Production Department is on its way of producing bottled calamansi juice as a healthy drink to be sold as an IGP of the campus. Calamansi juice is a beverage with health potential as a good source of vitamin C, however its acceptability is muted from other citrus juices such as lemon, lime and oranges. This study is anchored on sustainable development goal (SDG) 3: Good Health and Well-being since it aims to develop a nutritious drink with less sugar. The development of a ready-to-drink calamansi juice may answer the need of enhancing the nutritional value of calamansi juice and at the same time promoting calamansi at par with the other types of citrus fruits that would help in strengthening our calamansi industry.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The project intends to develop a ready-to-drink calamansi juice with low sugar content using stevia powder as sugar replacement. CapSU Pontevedra standard formulation of calamansi juice will be used as a starting point and will act as the control treatment. The amount/concentration of stevia powder will be varied in the experimental treatments. Concentrations of stevia powder will be patterned based on published data on the application of stevia on similar food matrix. Physico-chemical test, microbiological analysis, sensory evaluation and product's acceptability among treatments at different storage time will be assessed.

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(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives of the Study

Generally, the proposed study aims to develop a non-pasteurized, ready-to-drink calamansi juice with low sugar content.

Specifically, it aims to:

1. Determine the effect of stevia powder substitution on the physico-chemical properties of calamansi juice during storage;
2. Determine the effect of stevia powder substitution on the microbiological characteristics of calamansi juice during storage;
3. Determine the effect of stevia powder substitution on calamansi juice's sensory profile of the acceptability using consumers test;
4. Evaluate if there is a significant difference on the physico-chemical, microbial and sensory properties of calamansi juice among the treatments at different storage time; and
5. Evaluate the product's consumers acceptability.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

An increased interest in the application of stevia in food production has been observed every year according to Kobus-Moryson & Gramza-Michałowska (2015) because stevia answers the needs of consumers for a taste of sweetness without the excessive calorie intake and contributing nutritional benefits as a source of many substances. Due to its high appeal and commercial potential, further research on its application to various foods is needed to improve the taste of products sweetened with these compounds. The continued evaluation of these ingredients with regards to aspects such as the intensity, persistence of sweet taste and absence of other residual flavors is necessary in order to meet the demands of today's consumers and ensure their acceptance, preference and choice by the general public (González, Tapia, Pérez, Pallet, & Dornier, 2014).

Stevia leaves has already been applied to several fruit juices. In the work of Iqbal, Ayub, Khan, & Muhammad, (2020) different concentrations of diluted stevia extract with sweet orange juice and preserved with sodium benzoate (0.1%), potassium meta bisulphite (0.1%) and potassium sorbate (0.1%) at different ratios showed significant effect on the product's quality. In terms of its physico-chemical and organoleptic characteristic, the ratio of 3 mL stevia extract to 97 mL orange juice was the most acceptable treatment.

The physicochemical characteristics and sensory profile of mango nectar sweetened with different high intensity sweeteners throughout storage time was studied by Cadena et al.(2013). The mango nectar samples were sweetened with: acesulfame-K/sucralose/neotame blend (100:50:1), sucrose, stevia with 97% rebaudioside, neotame, sucralose and a thaumatin/sucralose blend (1:1) were carried out at Day zero, 60 days and 120 days of storage. The physicochemical analyses measured included color (L^* , a^* , b^*), pH, titratable acidity, soluble solids ($^{\circ}$ Brix) and ratio (Brix/titratable acidity). The sensory profile was studied using the Quantitative Descriptive Analysis (QDA). Result of the study showed that the pH values and titratable acidity of the mango nectars have no significant differences at

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$\alpha = 0.05$ between the different samples. Total soluble solid content ($^{\circ}\text{Brix}$) values did not change throughout the shelf life. With respect to its sensory profile, sucralose was judged to be the best substitute for sucrose when compared with the other high intensity sweeteners at both zero time and after 120 days of storage. The sample sweetened with sucralose showed acceptance (mean at storage time 6.4) and sensory profile equal to control (sucrose). In addition, the sweeteners stevia with 97% rebaudioside did not show off-flavor and the thaumatin/sucralose blend (1:1) and also presented similar acceptance (6.16 at Day zero) and sensory profile in relation to the control.

Apricot nectar was also produced using stevia as sugar. Results of physicochemical properties showed that there were no significant differences in pH and TTA values between the nectars produced with stevia and those produced with or without sucrose, highlighting that the addition of stevia did not affect the acidity of the products. All the products are liked, but with a different level of pleasantness. The study highlighted that the apricot nectars with 0.07% stevia are characterized for sweet and liquorice aroma notes and received the same level of consumer acceptability of nectars produced with 10% sucrose (Reale et al., 2020).

Ready-to-serve (RTS), low calorie fruit beverages from jamun, mango, pomegranate, pineapple, and purple grapes were also produced using optimized quantities of stevia aqueous extract and a commercial stevioside powder as replacement for sugar. The RTS beverages were analysed for physico-chemical and sensory parameters during storage for six months. The RTS beverages were well accepted by sensory panelists and found that the beverages were stable up to four months of storage, during which changes in physico-chemical parameters and sensory scores were found to be minimal (Balaswamy, Rao, Rao, & Nagender, 2014).

Other studies on the application of stevia was on fruit base milkshake (Alizadeh, 2014), apple preserve (Iqbal et al., 2020), guava drink (Jubeen et al., 2019) and functional herbal-fruit beverage (Skapska et al., 2020).

A Calamansi (*Citrofortunella microcarpa*) blend powdered beverage with patent application No. PH22016000218U has been published in Espacenet. The powdered beverage claimed to be composed of 44.60 percent white refined sugar, 30 percent Confectioner's sugar, 25 percent calamansi powder, 0.10 percent iodized salt and 0.30 percent citric acid.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested).

Most studies exploring the application of stevia in foods centers on its use as substitute for sugar/sucrose to create low-sugar food or beverages. The amount/concentration of the alternative sweetener (stevia) is commonly used as the independent variable while parameters that are usually included to measure its effect on the developed product are physicochemical characteristics such as total soluble solids (TSS, in $^{\circ}\text{Bx}$), titratable acidity (TA), pH, color (using colorimeter/chromameter), viscosity, TSS to TA ratio and etc. Microbial quality are evaluated in terms of yeast and molds counts, total plate count, total mesophilic count, LAB count, Enterobacteriaceae, enterococci and spore-forming bacteria. Finally, sensory evaluation, descriptive sensory profiling, consumer's acceptability testing are also being done to evaluate the product with stevia a sugar replacement.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

a. The protocol of the study is shown in the figure below:

Experimental Design: Randomized Complete Block Design (RCBD)

Treatments: A - control (standard formulation)

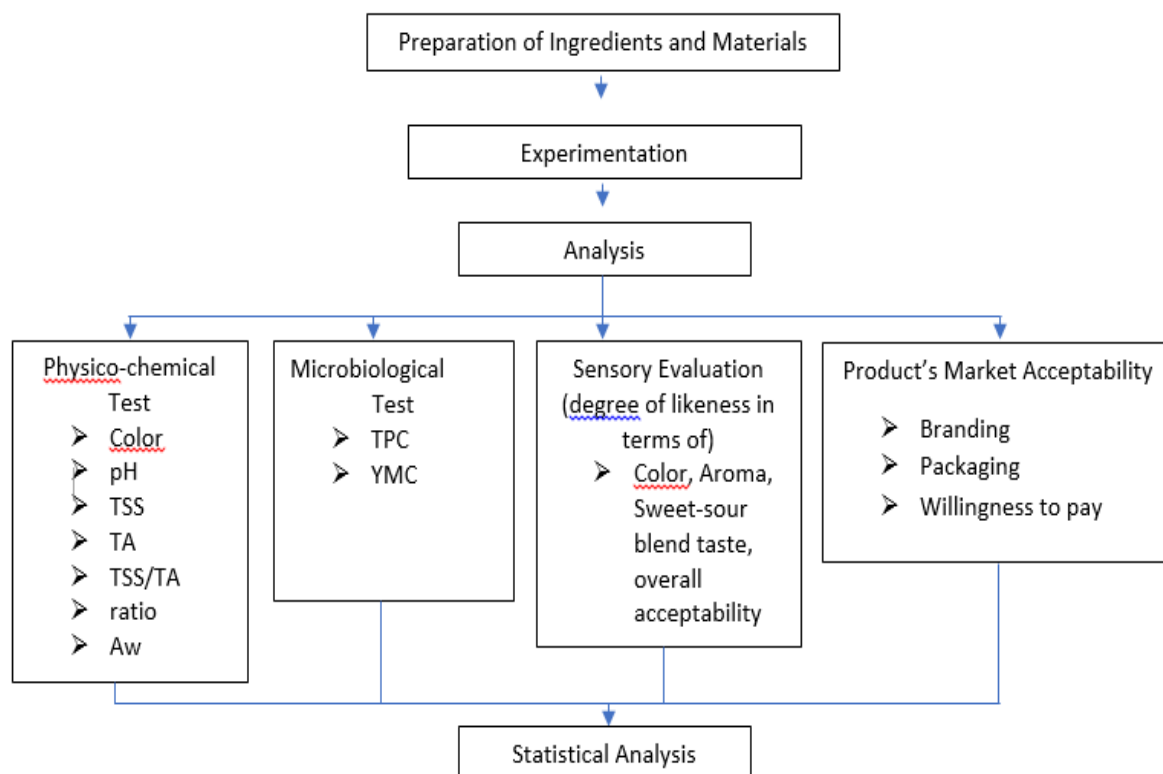
B – 10% substitution (1.6 g stevia powder: 100 g sugar)

C – 20% substitution (3.2 g stevia powder: 200 g sugar)

D – 40% substitution (6.4 g stevia powder: 400 g sugar)

E – 80% substitution (12.8 g stevia powder: 800 g sugar)

Block : Shelf life (in days) – 0, 3, 5 and 7 days old calamansi juice



b. Experimental Lay-out

Table 1. The proposed experimental lay-out.

TREATMENT	SHELFLIFE (in days)			
	0	3	5	7
A – Control (std. formulation)				
B – 10% substitution				
C – 20% substitution				
D – 40% substitution				

Note: Experiment will be done in triplicate

Research Activity	July				August				September			
	1	2	3	4	1	2	3	4	1	2	3	4
Refinement of proposal												
Purchase and preparation of materials												
Experimentation												
Data gathering												

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- Chromameter/Colorimeter		15,000
- plate count agar		2,500
- yeast and mold count agar		2,500
- reagents (NaOH, Na ₂ S ₂ O ₃)		5,000
- bond paper (6 reams)		1,500
- Miscellaneous (for incidental expenses)		5,000
- other office supplies (ink, ballpens, pencils)		5,000
C. CAPITAL OUTLAY		
Total		105,500.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

A CALAMANSI (Citrofortunella microcarpa) BLEND POWDERED BEVERAGE (2017) Available in <https://worldwide.espacenet.com/patent/search/family/058412754/publication/PH22016000218U1?q=calamansi%20blend%20CSU-DOST02>

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Morenga, L., Mallard, S., & Mann, J. (2013). Dietary sugars and body weight: systematic review and meta-analyses of randomised controlled trials and cohort studies. *BMJ* 2013, 346, e7492. <https://doi.org/10.1136/bmj.e7492>

Skapska, S., Marszalek, K., Wozniak, L., Szczepanska, J., Danielczuk, J., & Zawada, K. (2020). *Functional Fruit-Herbal Beverages*. 1–16.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

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